# Tarea lección 7.1

9) $\alpha = \underline{77°33'}$ $\quad a = 140$

$\beta = \underline{49°7'}$ $\quad b = \underline{108}$

$\gamma = 53°20'$ $\quad c = 115$

$$\frac{\operatorname{sen} 53°20'}{115} = \frac{\operatorname{sen}\alpha}{140} = \alpha = \operatorname{sen}^{-1}\left(\frac{140 \operatorname{sen} 53°20'}{115}\right) = \underline{77°33'}$$

$$180 - 77°33' - 53°20' = 49°7' = \beta$$

$$\frac{\operatorname{sen} 53°20'}{115} = \frac{\operatorname{sen} 49°7'}{b} = b = \frac{115 \operatorname{sen} 49°7'}{\operatorname{sen} 53°20'} = \underline{108.39}$$

15) $\alpha = \underline{25.993°}$ $\quad a = \underline{0.146}$

$\beta = 121.624°$ $\quad b = 0.283$

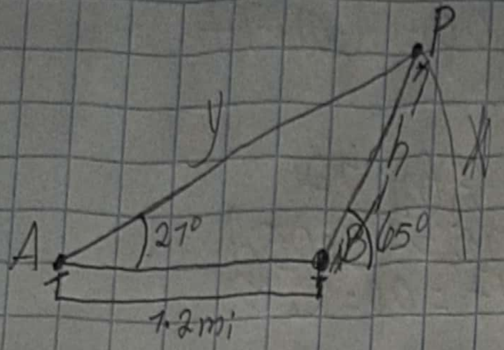
$\gamma = \underline{32.383°}$ $\quad c = 0.178$

$$\frac{\operatorname{sen} 121.624°}{0.283} = \frac{\operatorname{sen}\gamma}{0.178} = \gamma = \operatorname{sen}^{-1}\left(\frac{0.178 \operatorname{sen} 121.624°}{0.283}\right) = 32.383°$$

$$180 - 121.624 - 32.383 = 25.993$$

$$\frac{\operatorname{sen} 121.624°}{0.283} = \frac{\operatorname{sen} 25.993}{a} = a = \frac{0.283 \operatorname{sen} 25.993}{\operatorname{sen} 121.624} = 0.146$$

19)




$\tan 65° = \frac{h}{x} = h = \tan 65° x$

$\tan 27° = \frac{h}{1.2 + x} = h = \tan 27° (1.2 + x)$

$\tan 65 x = 1.2 \tan 27 + \tan 27 x$

$\tan 65 x - \tan 27 x = 1.2 \tan 27$

$x = \frac{1.2 \tan 27}{\tan 65 - \tan 27} = 0.26 mi$

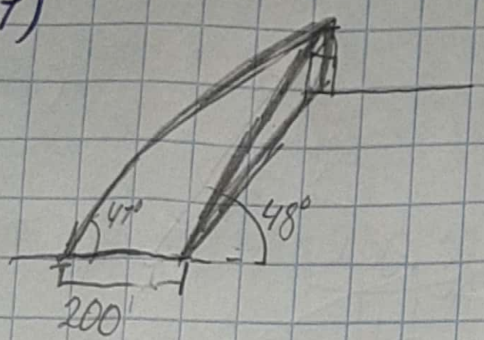
$h = x \tan 65 = 0.60$

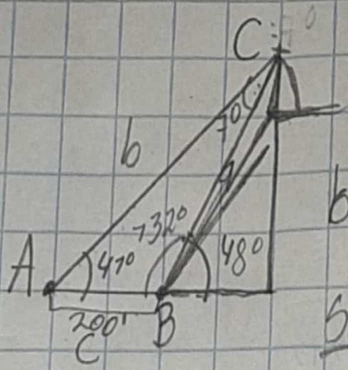
$\text{sen } 27 = \frac{h}{y} = y = \frac{h}{\text{sen } 27} = 1.60$

a) La distancia entre a y p es de 1.60 mi

b) La altura de la montaña es de 0.60 mi

27)




$\frac{\text{sen } 132}{b} = \frac{\text{sen } 1}{200}$

$b = 1,219.58$

$\frac{\text{sen } 47}{a} = \frac{\text{sen } 132°}{1,219.58}$

$a = 1,076.66$

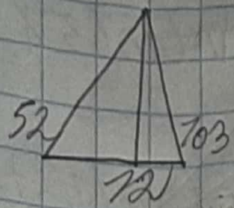
$B = 76°$
$A = 122°$
$C = 42°$

$\frac{\text{sen } 122}{1076.66} = \frac{\text{sen } 76}{b}$

$b = 350$

La altura de la catedral es de 350 pies.

29)

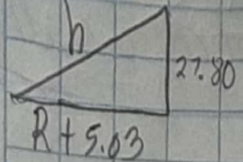



$\tan 52 = \frac{OP}{72 + A}$ $\quad \therefore op = A\tan 77$

$\tan 52(72 + A) = A\tan 77$

$72\tan 52 + A\tan 52 = 3.05A$

$A = 5.03 \cdot \tan 77 = op = 21.80$




$h = \sqrt{(72 + 103)^2 + 21.80^2}$ $\quad B = \frac{78 \cdot 21.80}{2}$

$h = 78.66$ $\quad V = \frac{1}{3}(796.2)(75.66)$

$V = 7024.76$